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COS30045 Data Visualisation

Class 2

Agenda

- Data processing with KNIME
- Answering questions with data



Time goes faster and you have more fun with friends!

- introduce your self to someone you don't know while waiting for class to start!

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Exercise 1.0 Data Processing with KNIME (see Canvas)

LAST WEEK

Scenario

Imagine you have been asked to create a set of visualisations that allows users to explore the energy consumption of televisions currently available on the Australian market.

Tasks

- Find the data set - DONE
- Open data set in KNIME - DONE
- Become familiar with data set - DONE
- Clean the data set - DONE
- Transform the data set - DONE



Exercise 2.0 Data exploration and Chart Types with KNIME (follow Canvas guide)

Overview

1. Answering questions with data
2. Create different charts with different types of data
3. Aggregate the data set
4. Mathematically transform data
5. Create new columns
6. Generate Charts
7. Use a pivot table to explore data

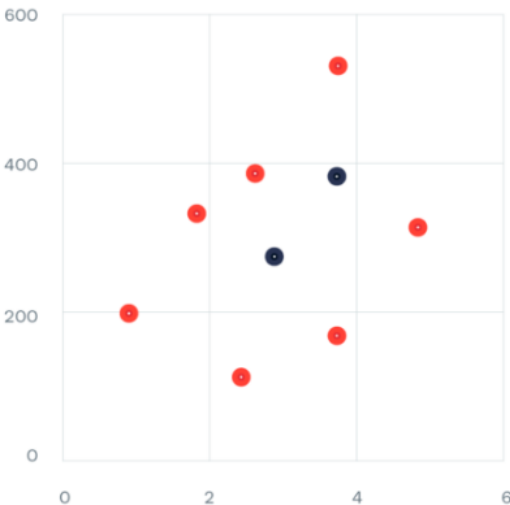


Common Chart Types

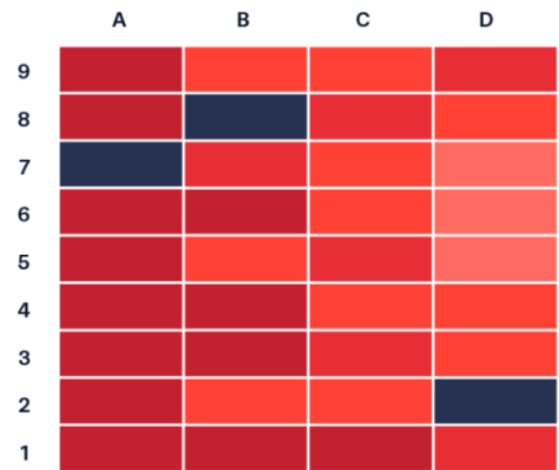
Line Graph



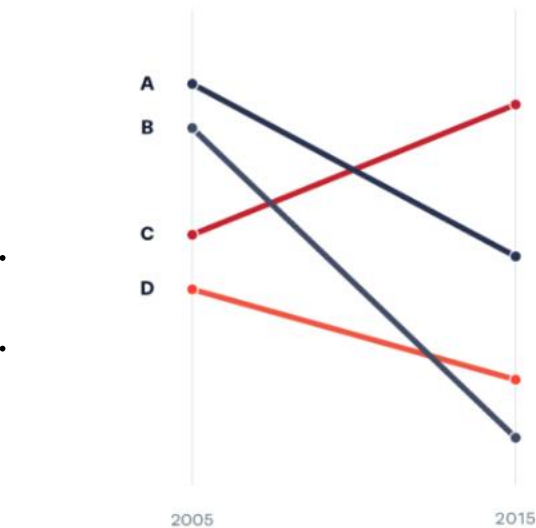
Scatter Plot



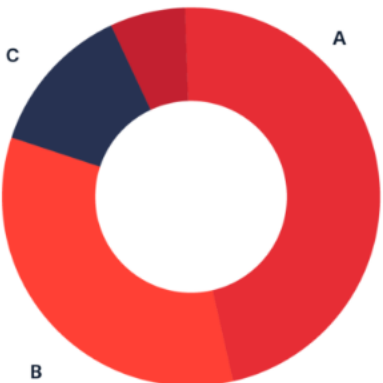
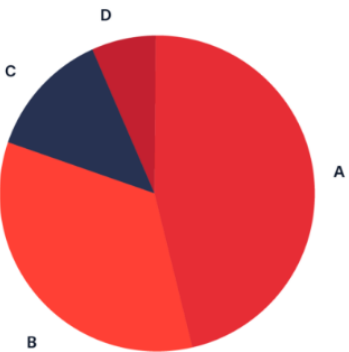
Heat Map



Slope Chart

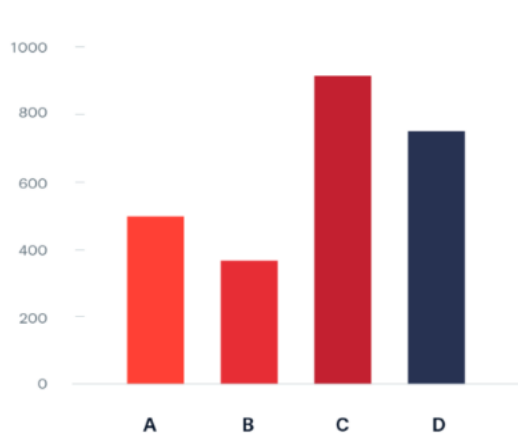


Pie Chart

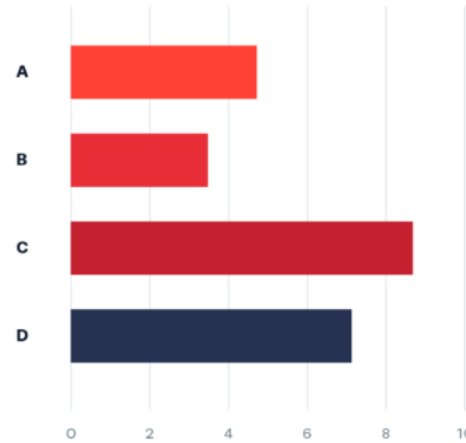


Common Chart Types

Bar Chart (Vertical)



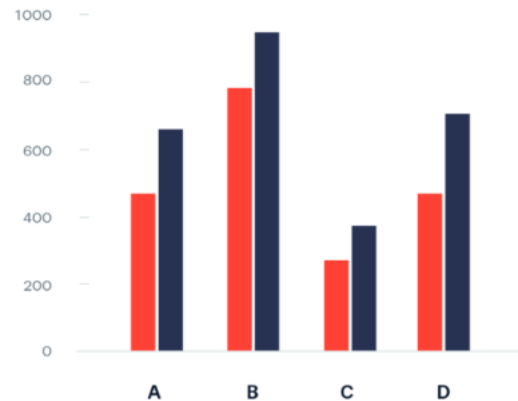
Bar Chart (Horizontal)



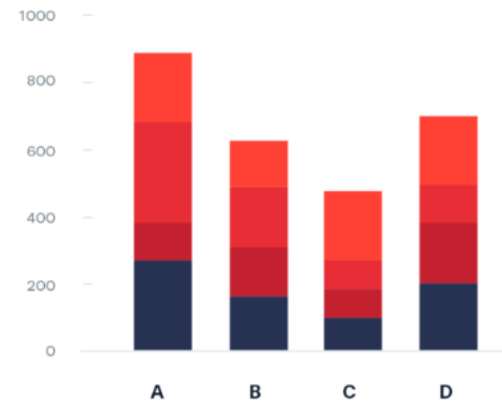
For list and description of chart use see

<https://datavizproject.com/>

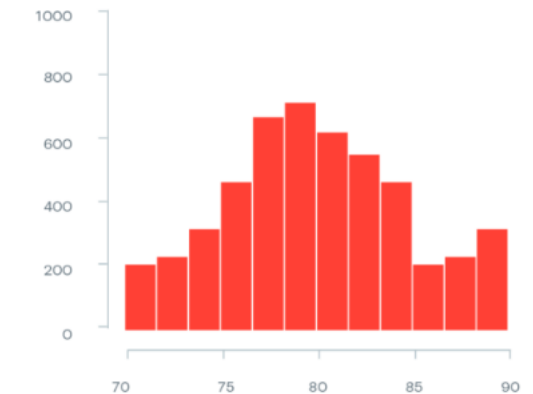
Grouped Bar Chart



Stacked Bar Chart



Histogram



<https://datavizproject.com/>

Exercise 2.0 Data exploration with KNIME (follow Canvas guide)

In this exercise you will use the same data set to:

- answer questions about the data
- create different types of charts using quantitative and qualitative data
- mathematically transform a data value
- create a new category column using an expression
- use a pivot table to explore data



Exercise 2.0 Data exploration with KNIME (follow Canvas guide)

1. What screen sizes are currently available, and which are the most frequent?
2. How is screen size and energy consumption related?
3. How is screen type and energy consumption related?

Other: Are there differences in power consumption between brands?

Any other questions you can think of that could be investigated with this data set?



Visualisation Critique (due Week 5)

Aim: Critique visualisation according to data visualisation guidelines

Due: Week 5 (just before Easter)

Structure

- Introduction
- Data Processing and Data Representation
- Visualisation Design
- Recommendations

Must be on specified visualisation
(see Canvas for this semester's visualisation)

Website being Critiqued
changes every semester
– Check Canvas for
details

Check requirements and bring
questions to class next week!

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Exercise Demonstration 1 (Week 4)

Be prepared to demonstrate:

Week 1 and Week 2:

- Data processing steps
- KNIME workflows
- Interpretation of TV data set (e.g., what the data means, how it can be used to answer questions etc)

Submission (night before your Class)

- Annotated KNIME export file (knar.knwf)
 - Must contain data
 - Include work done for Class Exercises
- MUST include your name in the file name

TVDataJamesMason.knar.knwf

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Exercise Demonstration 1

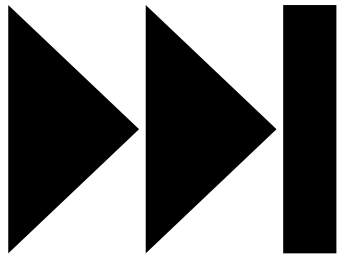
Face to face Q&A can be difficult for some people

Contact Convenor if you have concerns

Try not to stress about Demo!

It's not an exam, just a discussion about work you have done.





Next Week

Exercise 3: Communicating with Data

- Tell a story about the data!
- Present story in web site

